

Relation Algebra

§9. $\frac{R}{S}$

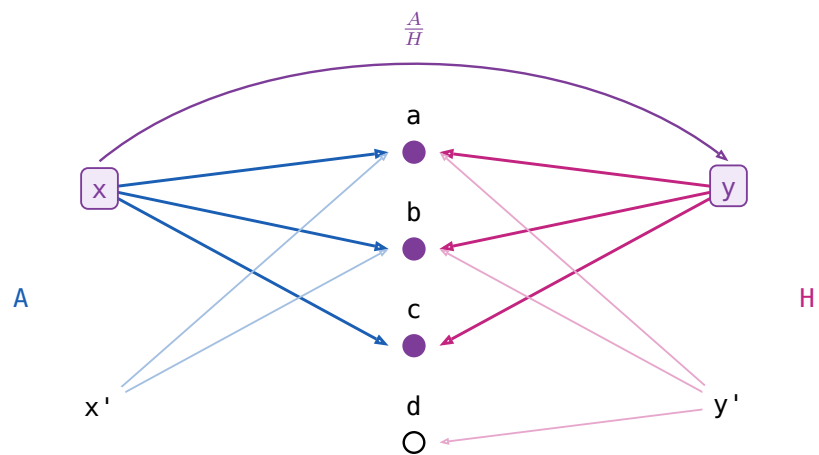
$$R /_s S \triangleq (R / S) \cap (S / R)^\circ \text{ what } R \text{ sends } x \text{ to and nothing else is what } S \text{ sends } y \text{ to. In Rel } x \text{ and } y \text{ has the same image: } \forall p. (x \text{ R } p \iff y \text{ S } p) \quad (9a)$$

$x \text{ (A/H) } y$ — x admires everyone y hates

$x \ ((H/A)^{\circ}) \ y$ — x admires only people y hates

$$x \ ((A/H) \cap (H/A)^\circ) \ y \text{ --- } x \text{ admires only and all whom } y \text{ hates}$$

x admires exactly whom y hates: / is **all of**, $\frac{R}{S}$ is **only all of**.



law	the reading
$X \sqsubseteq R /_s S \iff X S \sqsubseteq R \wedge X^\circ R \sqsubseteq S$ X sits inside the matching exactly when it carries S back inside R and, reversed, carries R back inside S .	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$(R /_s S)^\circ = S /_s R$ reversing the matching swaps the two relations.	Matching is symmetric.
$(R /_s S) (S /_s W) \sqsubseteq R /_s W$ a match from R to S followed by one from S to W is a match from R to W .	And transitive.
$(R /_s S) S \sqsubseteq R$ following the matching by S lands inside R .	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S}S = \text{Dom}(\frac{R}{S})R$
$(R /_s R) R = R$ matching R against itself and then following R gives R back.	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people $y=x$ always qualifies ($1 \sqsubseteq R \circ R$ below)
$1 \sqsubseteq R /_s R$ every x matches itself.	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$(R /_s R) (R /_s R) = R /_s R$ matching twice matches no more pairs than matching once.	So the relation admires the same people is an equivalence relation.
$\text{Symmetric}(X) \implies X \sqsubseteq R /_s R \iff X R \sqsubseteq R$ a symmetric X sits inside the matching exactly when following it by R adds nothing to R .	The largest symmetric arrow that leaves R alone.
$\text{simplePart}(R) \triangleq R /_s 1$ matching R against the identity keeps the x whose image is a single point.	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1=R$.
$\text{dom}(R /_s S) = 1 \cap (R /_s S) (S /_s R)$ the x that match something are those the two one-sided divisions take round the loop back to x .	Its domain is the domain of simplicity of R .